

Zusammenfassung databases

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CONTENTS

Contents	1
1 Intro	2
1.1 Organization	2
1.2 History of databases	2
2 SQL	2
2.1 Normalization	3
2.2 Transactions	3
3 Relational model	3
3.1 Identifying	4
3.2 Constraining	4
4 Relational algebra	4
4.1 Joining tables	4

1 Intro

1.1 Organization

Practicum(Couples):

- Opgave 1: Case description, modeling, schema design
- Opgave 2: Reconsider your design using AI

Assignments(3x) mandatory:

- Indication for questions final test
- Strong correlation between test results and attention in the lectures

What is a database? Storing data in tables, using a relational approach. Every entry has it's own primary key used to uniquely identify a table row. For example: A booknumber or a social security number.

Databases are omnipresent and the technology is directly used in the software project. Furthermore, it forms the backbone of most information systems. Studying database technology provides insight in general computer science principles:

- Layered software architecture
- Mathematical modeling
- Application of predicate logic and set theory

1.2 History of databases

In the eighties, the relational data model received widespread commercial attention.

SQL became the standard in 1986

SQL92/SQL2, SQL all follow ANSI standards

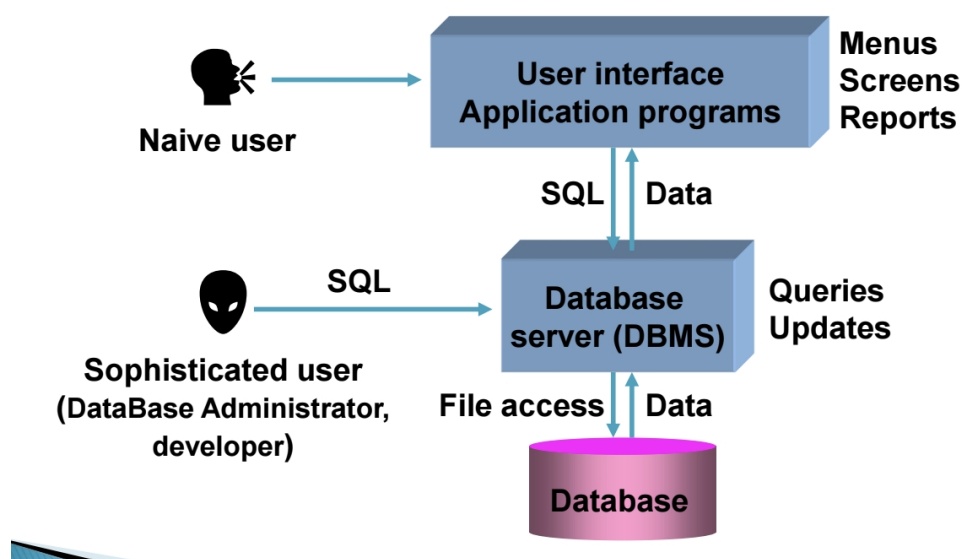
2 SQL

SQL is a Declarative language.

```
SELECT Name
FROM Book, Loan, Reader
Where Book.Title = 'Fun-Fishing'
      AND Book.Bno = Loan.Bno
      AND Loan.Rno = Reader.Rno
```

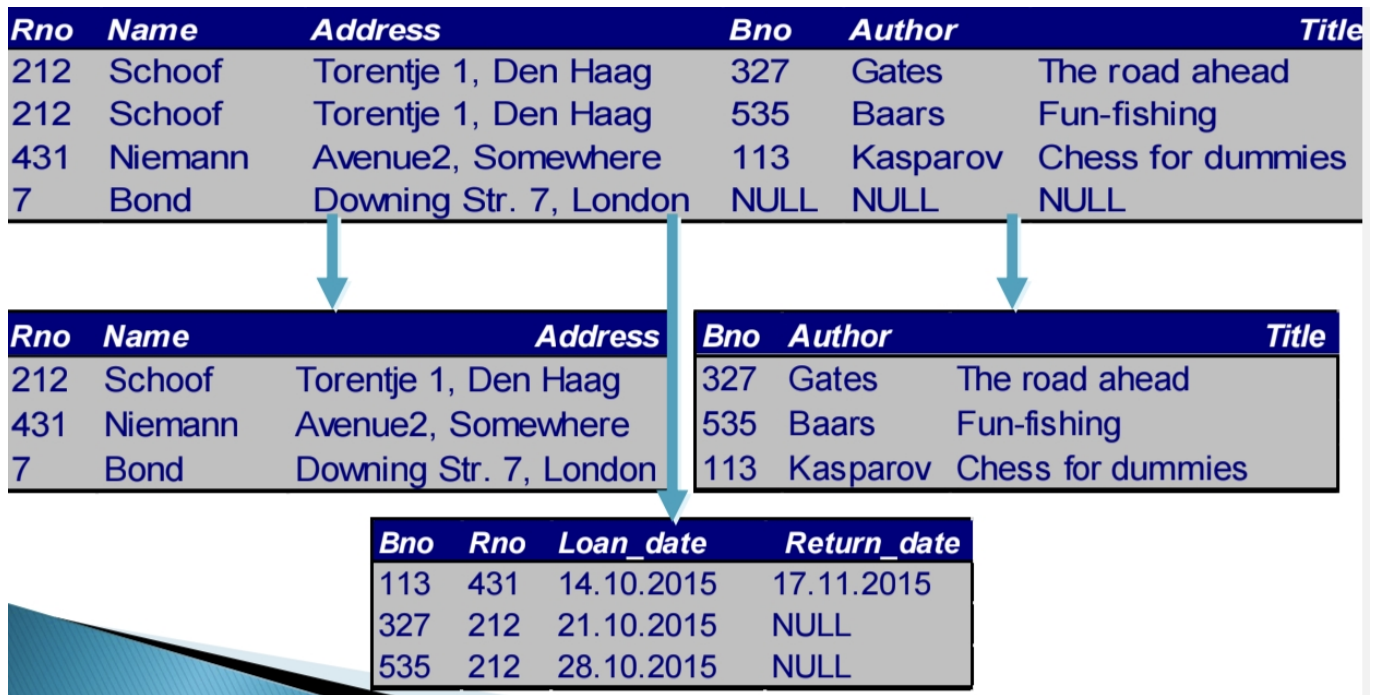
The example above returns a list of people to whom the book 'Fun-Fishing' is currently being loaned. From the table Book, it selects every entry or row with the name 'Fun-Fishing' and checks whether the booknumber matches a booknumber in the table Loan. It then fetches the reader from the table reader based on the reader number that matches the entry in the loan table.

A user can fetch(= request) data from a database thru a User interface application program, which sends SQL queries(= statements/prompt) to the database server. A developer or administrator can update the database using queries. The database server itself handles storing and accessing the data from the database. As is illustrated in the diagram below:



2.1 Normalization

To improve readability, manageability and adequate representation (avoiding NULLs) data usually gets separated in different tables. If we take a look at our library example, by having a different table for: Books, Loans and Readers, we can just delete an entry in the table loans, when a book is returned. This has the advantage that when a Reader isn't reading a book, there are not a bunch of NULL entries in our data table. This also solves an interesting problem when allowing a reader to loan multiple books, using multiple tables, a new entry in Loans can be made. Instead of some complicated logic using arrays in the general table. The image below depicts an example when looking at our library example:



2.2 Transactions

When working with transactions, it is important to either finish everything or nothing. When withdrawing cash for example, it would be suboptimal for the bank if an ATM prints out your cash, but then a crash happens and your balance doesn't get updated. It is either all or nothing.

Another problem that might occur is concurrency, when two queries are being executed at the same time, there might happen some unexpected results. Using this can be solved.

3 Relational model

Take the following SQL Table for example: The of this Relation is Movie(movid, title, year, rating)

Movie			
movid	title	year	rating
tt0469494	There Will Be Blood	2007	8,1
tt0086879	Amadeus	1984	8,4
tt0102926	The Silence of the Lambs	1991	8,6
tt0110413	Léon	1994	8,7
tt0078788	Apocalypse Now	1979	8,5

There are five Tuples, every entry or row with the corresponding row number.

This Schema has four Attributes, namely: movid, title, year, rating.

There are four columns, defined by Attributes, each containing five values.

This makes it so that the Degree of the Relation Movie is four.

3.1 Identifying

Each SQL table has one Primary key, this key is used to uniquely identify a row or in this case a movie. Note however that Tuples are also identified by (movid, title), (movid, year) or any combination containing a key.

A Super key is a attribute or combination of Attributes that can uniquely identify every entry. In the example above, this would be movid. However, since in the example above every field can be used as a Super key, it is important to chose your Primary key wisely, since in the future there will be more movies with the same year or rating.

The Primary key is chosen from a set of Candidate keys, which are minimalistic super keys. Example: (movid, title) is a Super key since it can identify all entry, however it is not a candidate key, since (movid) can also identify every entry.

3.2 Constraining

Take a look at the following creation of a table:

```
CREATE TABLE Book (
  bookid          integer          not null ,
  title           varchar(100)     not null ,
  author          varchar(30)      not null ,
  price           float ,
  date_of_purchase date ,
  publisher_id     varchar (6) ,
  CONSTRAINT Book_pk PRIMARY KEY (bookid) ,
  CONSTRAINT Book_fk_Publisher FOREIGN KEY (publisher_id)
    REFERENCES Publisher(publisher_id)
);
```

Here, CONSTRAINT Book_pk, create a constraint named Book_pk that says: bookid is the primary key.

CONSTRAINT Book_fk_Publisher says that the attribute publisher_id serves as a foreign key for the table Publisher

4 Relational algebra

The Unary operator σ , is used for a selection with complex predicates. To notation is as follows:

$$\sigma_{birth_year < 1990 \wedge gender = 'female'}(Actor) \quad \text{or} \quad \sigma_{Country = 'Netherlands' \vee Country = 'Belgium'}(Actor)$$

Here, Actor denotes the name of the relation or table.

Another unary operator π , is used for projection. The following statement will return the title and year columns from our movies table:

$$\pi_{title, year}(Movie)$$

Following this syntax a composition of projection and selection can be constructed:

$$\pi_{name, birth_year}(\sigma_{birth_year < 1960}(Actor))$$

The binary operators $\cup$, $\cap$, and $\times$ for the cartesian product all behave in a similar manner, where the result of the cartesian product is just a bigger table containing all the possible outcomes like so:

```
var Cartesianproduct = [];
for(var rowA in TableA) {
  for(var rowB in TableB){
    Cartesianproduct.add(rowA + rowB)
  }
}
```

4.1 Joining tables

Using the binary operator natural join $\bowtie$, two sets can be combined using the default matching condition. A Foreign Key can help you with what to expect. From each row where the foreign key matches that of the other table, these row's are combined together. When row A from table 1 matches with row B & C from table 2, the record A B & A C are added to the result.

The binary operator theta-join $\bowtie_{\theta}$ behaves mostly the same way as natural join, however θ defines a matching condition. In the image below this is illustrated: In this example, the rows where the attribute A from the relation R matches the attribute C from relation S and the attribute B from relation R is bigger than attribute D from relation S are joined.

R		S	
A	B	C	D
a	11	b	55
b	43	c	31
c	37	b	21

$R \bowtie_{\theta} S$			
A	B	C	D
b	43	b	21
c	37	c	31

$$\theta : (R.A = S.C) \wedge (R.B > S.D)$$

Glossary

ANSI American National Standards Institute. 2

Attributes Fields or properties. 3, 4

Candidate keys A set of minimalistic super keys, meaning that there is not a super key that is build out of less attributes. 4

Declarative Denoting high-level programming languages which can be used to solve problems without requiring the programmer to specify an exact procedure to be followed. 2

Degree Amount of properties a table or relation has. 3

Foreign Key A set of attributes, that servers as a candidate key in another relation. 4

Primary key Chosen from the candidate keys. 4

Relation Different fields linked together, such as: movid x title x year x rating. Often displayed in tables when using SQL. 3

Schema An SQL Table or Relation. 3

SQL Sequel query language. 2

Super key Attribute or combination of attributes that can uniquely define a relation. 4

Tuples A row of objects, often denoted using (x, y). 3, 4